

# AI-powered summaries and responses from PDFs

## Overview

This project aims to develop an intelligent system that reads and understands PDF documents—specifically those containing company profiles, manuals, reports, schedules, product catalogs, and contact lists. Using AI-powered natural language processing (NLP), the system will extract key information, generate summaries, and provide accurate responses to user queries based solely on the content of the uploaded documents.

## Objectives

- To enable quick retrieval of information from extensive company documents.
- To generate concise summaries of lengthy PDFs, such as manuals and reports.
- To provide an interactive Q&A interface where users can ask questions and receive context-aware answers sourced directly from the PDFs.
- To improve document accessibility for employees, partners, and customers.

## Key Features

- **PDF Upload and Parsing:** Upload one or multiple PDF documents for processing.
- **Content Categorization:** Automatically classify content (e.g., manuals, reports, schedules).
- **Summarization Engine:** Generate AI-driven summaries for quick understanding.
- **AI Chat Interface:** Interactive chat powered by AI to answer questions with references from the PDFs.
- **Search and Highlight:** Keyword-based search with highlighted results in original documents.

- **Access Control:** Role-based permissions to control who can view, search, or ask about specific documents.

## Target Users

- **Company staff** looking to quickly access internal knowledge.
- **HR and onboarding teams** providing manuals and company info.
- **Sales and support teams** needing product specs and contacts.
- **Executives** reviewing reports and schedules efficiently.
- **Accounting and finance teams** managing and analyzing financial reports, contracts, and cost summaries

## Benefits

- Save time by avoiding manual reading of long documents.
- Reduce repetitive internal queries by offering instant answers.
- Improve knowledge accessibility and decision-making across departments.
- Support remote or distributed teams with centralized document intelligence.

## Technology Stack

### Frontend:

- React.js with Tailwind CSS

### Backend:

- Laravel 11 (PHP)
- REST API for communication between frontend and backend

### Document Processing & AI:

- PDF.js or PyMuPDF for PDF parsing
- Llama 3.3 AI Model for summarization and question-answering

### Database:

- MySQL for storing metadata and user activity

**Hosting & Deployment:**

- Cloud Hosting via Azure
- SSL-secured deployment with basic firewall configurations

**Project Timeline**

| Phase   | Description                                                  | Deliverables                                                                  | Duration  |
|---------|--------------------------------------------------------------|-------------------------------------------------------------------------------|-----------|
| Phase 1 | Requirement Analysis & System Design                         | System requirements document, UI/UX wireframes, technical architecture plan   | 1 week    |
| Phase 2 | Core Functionality Development (PDF parsing, AI integration) | Upload & Parse PDF, AI summarization engine, basic Q&A chatbot                | 2 weeks   |
| Phase 3 | Frontend and Backend Integration                             | Functional UI with chat, document viewer, and basic admin/user authentication | 1 week    |
| Phase 4 | Internal Testing & Debugging                                 | Bug fixes, performance optimizations, initial user testing reports            | 1 week    |
| Phase 5 | Feedback Implementation & Final Polishing                    | UI/UX refinements, improved responses, optimized data access                  | 1–2 weeks |